OPERATING SYSTEMS: FUNDAMENTALS ASSIGNMENT

Course: Computer Science 301 | **Total Marks:** 100 | **Due Date:** As per Academic Calendar

ASSIGNMENT GUIDELINES

- **Word Count:** Each answer must be approximately **100 words**. Be concise, clear, and focused on the core concepts.
- **Originality:** All responses must be your own work. Plagiarism or AI-generated text will result in a zero mark.
- **Formatting:** Clearly number your answers corresponding to the questions below.
- **Submission:** Submit your final response file in PDF format via the online student portal.

1. Explain the primary purpose of an operating system and its two main roles from a user and system perspective. [4 Marks]

2. What is the difference between a process and a program? Discuss how one transforms into the other. [4 Marks]

3. Describe the main differences between a microkernel and a monolithic kernel architecture. [4 Marks]

4. Explain the concept of a dual-mode operation (User Mode vs. Kernel Mode) in an operating system and why it is crucial for system security. [4 Marks]

5. What is a system call? Explain the sequence of events that occurs when a user application requests a service via a system call. [4 Marks]

6. Define the term "Context Switching" and list the specific components stored in the Process Control Block (PCB) during this event. [4 Marks]

7. Contrast the characteristics and primary functions of short-term, medium-term, and long-term CPU schedulers. [4 Marks]

8. Explain the phenomenon of 'starvation' in CPU scheduling algorithms and describe how the technique of 'aging' resolves it. [4 Marks]

9. Differentiate between preemptive and non-preemptive scheduling algorithms, providing one distinct example of each. [4 Marks]

10. What are the benefits and drawbacks of multi-threading over multi-processing for modern concurrent application development? [4 Marks]

11. Define the 'Critical Section Problem' and explain the three essential requirements that any valid solution must satisfy. [4 Marks]

12. What is a race condition? Provide a brief conceptual example of how a race condition can lead to inconsistent data. [4 Marks]

13. Explain how binary semaphores differ from counting semaphores, and briefly describe a scenario where each would be utilized. [4 Marks]

14. Identify and describe the four necessary conditions that must hold simultaneously for a system deadlock to occur. [4 Marks]

15. Contrast the strategies of deadlock avoidance (such as the Banker's Algorithm) with deadlock prevention. [4 Marks]

16. Explain the difference between logical addresses and physical addresses, and clarify which component handles their runtime mapping. [4 Marks]

17. Discuss the difference between internal and external fragmentation in memory management, citing which memory allocation schemes suffer from each. [4 Marks]

18. Explain the core mechanics of paging and how a Page Table is leveraged to locate frames in physical memory. [4 Marks]

19. What is virtual memory? Explain how it benefits both developers and the system's physical hardware constraints. [4 Marks]

20. Describe the exact sequence of events that takes place when a 'Page Fault' is triggered by the hardware. [4 Marks]

21. Explain Belady's Anomaly and name the specific page replacement algorithm that is known to suffer from it. [4 Marks]

22. Define 'Thrashing' in the context of memory management and describe its immediate impact on overall CPU utilization. [4 Marks]

23. What is an Virtual File System (VFS), and how does it allow an operating system to support multiple different physical file systems seamlessly? [4 Marks]

24. Compare indexed allocation with linked allocation methods for storing files on a disk, highlighting one major trade-off. [4 Marks]

25. Explain the concept of Direct Memory Access (DMA) and why it is vastly superior to programmed I/O for large data transfers. [4 Marks]